

Algorithmic Trading Execution Engine

Domain: Algorithmic Trading | Statutory Architecture

Parameter	Evaluation Result	Governance & Statutory Impact
MoSCoW Baseline	Must Have	Ensures compliance with regulatory requirements and maintains high performance standards.
Risk Classification	MEDIUM (61/100)	Moderate operational tier. Asynchronous reconciliation, compliance reporting, or account aggregator data streaming.
Total Requirements	38 Modular Requirements (Min 30)	Decomposed across Frontend, Backend, Auth, Security, Database Residency, and Resilience

1. System Scope & Stakeholder Requirement Statement

"Algorithmic trading execution engine with low latency FIX protocol, colocation in NSE, and real-time risk checks."

2. Modular Software Requirements Specification (38 Detailed Points)

Pillar: Frontend & Mobile UI/UX (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-FE-01	Real-Time Level-2/3 Market Depth & Tick Stream Interface Deliver sub-50ms streaming market depth and order book visualizer for Algorithmic Trading Execution Engine displaying top 20 bids/asks with volume delta indicators.	Must Have	Regulation: SEBI CSCRF Framework & Exchange Colocation Guidelines Acceptance: Renders WebSocket market depth updates with zero UI frame drops at 50 updates/sec.
REQ-FE-02	High-Speed Order Entry Ticket & Hotkey Execution Matrix Provide ultra-fast order placement dialog supporting Limit, Market, Stop-Loss, and Bracket orders with configurable single-stroke hotkeys.	Must Have	Regulation: BSE/NSE Trading Regulations & Algorithmic Order Norms Acceptance: Executes client-side form validation and submits order payload in <10ms.
REQ-FE-03	Algorithmic Strategy Management & Execution Console Deliver a dedicated interface to deploy, parameterize, pause, and monitor automated trading strategies (TWAP, VWAP, Arbitrage) in real time.	Must Have	Regulation: SEBI Guidelines on Algorithmic Trading Systems Acceptance: Displays live execution slippage, executed lots, and strategy P&L; in real time.
REQ-FE-04	Emergency Global Kill-Switch & Position Liquidation Control Enforce prominent, fail-safe panic button that cancels all working orders across all exchanges and squares off open positions within 500ms.	Must Have	Regulation: SEBI Circular on Risk Containment & Circuit Breakers Acceptance: Dispatches atomic cancel-all requests to NSE/BSE and verifies acknowledgement within 500ms.
REQ-FE-05	Real-Time Margin Utilization & P&L; Analytics Dashboard Provide real-time visualization of Initial Margin, Extreme Loss Margin (ELM), MTM P&L;, and peak margin headroom with automated margin call alerts.	Should Have	Regulation: SEBI Peak Margin Regulations & Exchange RMS Directives Acceptance: Recalculates margin utilization on every tick; alerts trader when threshold exceeds 80%.

Pillar: Backend Architecture & Trading Engine (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-BE-01	Low-Latency FIX 4.4/5.0 Protocol Exchange Gateway Deploy high-performance FIX protocol engine connecting to BSE and NSE matching engines with automated heartbeat, sequence reset, and session recovery.	Must Have	Regulation: FIX Protocol Specification v4.4 & Exchange Architecture Standards Acceptance: Sustains message round-trip latency under 1 millisecond with zero dropped messages.
REQ-BE-02	Pre-Trade Risk Management (RMS) & Margin Verification Engine Enforce ultra-fast pre-trade risk checks verifying client margin, price band limits, order quantity thresholds, and single-order limits before dispatching to exchange.	Must Have	Regulation: SEBI Pre-Trade Risk Controls Mandate Acceptance: Evaluates 100% of orders in <150 microseconds; rejects orders breaching exposure limits.
REQ-BE-03	Multicast Market Data Feed Handler & Tick Normalizer Ingest raw UDP multicast market feeds from NSE (NOW/NEAT) and BSE, normalizing tick packets into structured internal market data buses.	Must Have	Regulation: NSE/BSE Market Data Feed Architecture Norms Acceptance: Parses 100,000 tick packets per second with packet loss rate < 0.0001%.
REQ-BE-04	Algorithmic Order Slicing & Smart Order Router (SOR) Implement smart order routing algorithms that dynamically split large parent orders into child slices across liquidity venues to minimize market impact.	Should Have	Regulation: SEBI Algorithmic Trading Best Practices Acceptance: Achieves average execution price within 0.05% of benchmark VWAP.
REQ-BE-05	Trade Drop Copy & Clearing Corporation Gateway Integration Transmit real-time drop-copy execution reports to risk surveillance systems and clearing corporations (NSCCL/ICCL) for trade allocation.	Must Have	Regulation: Clearing Corporation Real-Time Allocation Guidelines Acceptance: Dispatches execution reports within 50ms of exchange fill confirmation.

Pillar: Authentication, Identity & Access Governance (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-AUTH-01	Two-Factor Authentication (2FA) with Biometric & OTP Verification Mandate multi-factor authentication combining password/PIN with TOTP or biometric credentials on all trading terminals and mobile applications.	Must Have	Regulation: SEBI Circular on Mandatory 2FA for Trading Accounts Acceptance: Blocks access without successful 2FA; invalidates session after 3 failed attempts.
REQ-AUTH-02	SEBI Demat & UCC (Unique Client Code) Dynamic Validation Validate client UCC, PAN, and Demat account status against exchange master databases prior to permitting any order entry.	Must Have	Regulation: Exchange UCC Matching Directives & SEBI Master Circular Acceptance: Blocks order entry if UCC status is inactive, suspended, or unmapped in exchange records.
REQ-AUTH-03	Role-Based Access Control (Dealer, Risk Manager, Administrator) Enforce strict role separation ensuring dealers cannot modify risk parameters and risk managers cannot initiate proprietary trades.	Must Have	Regulation: SEBI Internal Controls Framework for Stock Brokers Acceptance: Zero privilege escalation; logs all administrative permission changes with dual signoff.
REQ-AUTH-04	Automated Trading Session Inactivity Lock & Device Binding Enforce device MAC/IMEI binding and automated terminal lock after 10 minutes of user inactivity.	Must Have	Regulation: SEBI Cyber Security Framework for Stock Brokers Acceptance: Locks workstation screen after 10m idle time; requires re-authentication to resume.
REQ-AUTH-05	Maker-Checker Authorization for High-Value Off-Market Transfers Require dual-authorization maker-checker workflow for off-market share transfers, margin limit enhancements, and collateral releases.	Must Have	Regulation: SEBI Supervisory Framework for Depository Participants Acceptance: Prohibits maker from self-approving; records digital signatures of both parties.

Pillar: Information Security & Statutory Compliance (7 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-SEC-01	Zero Trust Security Model Enforces a zero-trust security model where only authorized systems can connect to the trading engine.	Must Have	Regulation: SEBI CSCRf Circular SEBI/HO/MRD/TPD/P/CIR/2023/147 Acceptance: All connections to the trading engine require multi-factor authentication and encryption.
REQ-SEC-02	Data Encryption Encrypts all sensitive data at rest and in transit to protect against unauthorized access.	Must Have	Regulation: Statutory Compliance Directive Acceptance: Uses AES-256 bit encryption for all data storage and transmission.
REQ-SEC-03	SEBI CSCRf (Cyber Security & Cyber Resilience Framework) Compliance Implement comprehensive technical and procedural security controls meeting all statutory clauses of the SEBI CSCRf standard.	Must Have	Regulation: SEBI CSCRf Circular SEBI/HO/MRD/TPD/P/CIR/2023/147 Acceptance: Annual SOC2 Type II and CERT-In empaneled security audit certification.
REQ-SEC-04	TLS 1.3 Transport Security & Dedicated Leased Line Encryption Encrypt all order routing lines, market data streams, and API connections using TLS 1.3 with forward-secrecy cipher suites.	Must Have	Regulation: Exchange Network Security Mandates Acceptance: Rejects deprecated TLS versions (1.0, 1.1, 1.2); enforces AES-256-GCM.
REQ-SEC-05	Immutable WORM Forensic Audit Logging for Order Lifecycle Store every order event (placement, modification, cancellation, fill, reject) with microsecond NTP timestamps in Write-Once-Read-Many (WORM) storage.	Must Have	Regulation: SEBI Audit Trail Directives & Information Technology Act 2000 Acceptance: Logs stored in immutable tamper-evident storage for statutory 7-year retention period.
REQ-SEC-06	Real-Time Anti-Spoofing & Market Manipulation Surveillance Deploy algorithmic surveillance monitoring for spoofing, layering, circular trading, and front-running patterns with automated alert dispatch to the Compliance Officer.	Must Have	Regulation: SEBI Prohibition of Fraudulent and Unfair Trade Practices (PFUTP) Acceptance: Flags wash trades or rapid cancel-to-fill anomalies within 5 seconds.
REQ-SEC-07	Network Perimeter Web Application Firewall (WAF) & Anti-DDoS Protect internet-facing trading APIs against volumetric DDoS attacks and OWASP Top 10 vulnerabilities via adaptive rate limiting.	Should Have	Regulation: SEBI Capacity Planning & Cyber Security Directives Acceptance: Mitigates DDoS traffic up to 50 Gbps; enforces 200 req/min rate limit per IP.

Pillar: Database Architecture & Data Residency (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-DB-01	100% Indian Sovereign Data Residency in Colocation Centers Host all trading databases, market replay stores, and audit logs exclusively inside Tier-3/Tier-4 colocation data centers in Mumbai (Indian territory).	Must Have	Regulation: SEBI Data Localization Directives Acceptance: Zero persistent storage or transaction routing through foreign jurisdictions.
REQ-DB-02	High-Throughput Time-Series Tick Database Engine Deploy specialized time-series storage (e.g., ClickHouse / InfluxDB) optimized for sub-second ingestion and querying of billions of historical market ticks.	Must Have	Regulation: SEBI Historical Market Data Retention Norms Acceptance: Sustains 50,000 tick writes/second while supporting historical query execution in <200ms.

REQ-DB-03	ACID Financial Ledger for Client Margins & Trade Settlements Implement relational tables enforcing double-entry bookkeeping with serializable transaction isolation for all margin balances and trade settlements.	Must Have	Regulation: SEBI Client Fund Separation & Accounting Guidelines Acceptance: <i>Guarantees exact balance reconciliation; eliminates floating-point rounding discrepancies.</i>
REQ-DB-04	Statutory 7-Year Encrypted Archival & Retrieval Readiness Automate cold storage migration of trade contracts, contract notes, and client logs with scheduled cryptographic checksum verification.	Must Have	Regulation: Securities Contracts (Regulation) Rules 1957 Rule 15 Acceptance: <i>Produces complete statutory trade trail for any client within 4 hours of SEBI inquiry.</i>
REQ-DB-05	Multi-AZ Synchronous High Availability Clustering (RPO = 0) Deploy synchronous active-standby database replication across multiple availability zones with zero data loss failover guarantee.	Must Have	Regulation: SEBI Business Continuity Management Framework Acceptance: <i>Standby node promoted in <60 seconds upon primary node failure with zero committed trade loss.</i>

Pillar: Resilience, Scalability & Disaster Recovery (7 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-RES-01	Real-Time Risk Checks Performs real-time risk assessments on all executed trades to ensure market stability and prevent losses.	Must Have	Regulation: SEBI CSCRF Circular SEBI/HO/MRD/TPD/P/CIR/2023/147 Acceptance: <i>Risk checks completed within 1 second of each trade execution.</i>
REQ-RES-02	High Availability Maintains high availability by ensuring no more than 5 minutes downtime per year.	Should Have	Regulation: Statutory Compliance Directive Acceptance: <i>System uptime > 99.9% annually.</i>
REQ-RES-03	Sub-Millisecond Exchange Circuit Breakers & Order Throttling Incorporate algorithmic circuit breakers that instantly halt trading if market price deviation exceeds exchange price bands or account drawdown limits.	Must Have	Regulation: Exchange Market Wide Circuit Breaker Guidelines Acceptance: <i>Freezes order placement within 10 milliseconds of circuit trigger.</i>
REQ-RES-04	Active-Active Colocation Redundancy & Hot Standby Gateways Maintain hot-standby FIX connection sessions with exchange matching engines capable of instant session take-over upon primary gateway failure.	Must Have	Regulation: SEBI Infrastructure Resilience Norms Acceptance: <i>Gateway failover occurs in <2 seconds without losing in-flight order states.</i>
REQ-RES-05	Automated Cancel-on-Disconnect (COD) Exchange Safeguards Configure exchange-level Cancel-on-Disconnect flags ensuring all resting limit orders are automatically cancelled by the exchange upon connection drop.	Must Have	Regulation: BSE/NSE Trading Member Risk Management Guidelines Acceptance: <i>Exchange confirms 100% cancellation of working orders within 3 seconds of socket drop.</i>
REQ-RES-06	Geo-Redundant Disaster Recovery (DR) Site with RTO < 15m Operate a geographically distant Disaster Recovery site located in an alternate Indian seismic zone with biannual live mock trading cutover drills.	Must Have	Regulation: SEBI BCP/DR Mandate for Market Intermediaries Acceptance: <i>Workload cuts over to secondary DR site within 15 minutes during disaster scenario.</i>
REQ-RES-07	High-Throughput Peak Stress Testing (10,000 TPS Benchmark) Embed automated continuous load testing simulating extreme market volatility days (minimum 10,000 orders/sec) with zero message corruption.	Should Have	Regulation: SEBI Capacity Planning Directives for Stock Brokers Acceptance: <i>Maintains sub-2ms 99th percentile response latency under 10,000 TPS load.</i>

Pillar: Core Features (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-35	Low Latency FIX Protocol Support Supports the FIX 4.4/5.0 protocol for efficient trade executions with minimal latency.	Must Have	Regulation: SEBI CSCRF Circular SEBI/HO/MRD/TPD/P/CIR/2023/147 Acceptance: System can successfully execute trades over FIX protocol with <1ms latency.

Pillar: Integration (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-36	NSE Colocation Colocates the system with the National Stock Exchange (NSE) for immediate market updates and faster execution times.	Should Have	Regulation: Statutory Compliance Directive Acceptance: System receives market data updates within 10 ms of NSE.

Pillar: Testing (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-37	Pre-Production Testing Conducts thorough pre-production testing including formal traceability matrices and third-party audits.	Must Have	Regulation: SEBI CSCRF Circular SEBI/HO/MRD/TPD/P/CIR/2023/147 Acceptance: Passes all predefined tests and undergoes third-party CERT-In empaneled VAPT audit.

Pillar: Monitoring (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-38	Compliance Monitoring Monitors system operations and logs for compliance with regulatory requirements.	Should Have	Regulation: Statutory Compliance Directive Acceptance: Logs and reports generated every hour for review by internal auditors.

3. Regulatory & Statutory Grounding (Indian Banking Frameworks)

Framework	Enforcement Level	Statutory Directive / Citation
SEBI CSCR Framework (2023)	Mandatory Cyber Security & Resilience	SEBI/HO/MRD/TPD/P/CIR/2023/147
SEBI Algorithmic Trading Regulations	Algorithm Testing & RMS Compliance	SEBI Master Circular for Stock Brokers, Section 8
Exchange Colocation & Order Routing Guidelines	Network Latency & Architecture Norms	BSE & NSE Algorithmic Facility Guidelines
Digital Personal Data Protection Act, 2023	Trader Data Privacy & Security	DPDP Act 2023, Sections 6 & 8

4. Mandatory Security & Architectural Controls

Control ID	Title & Architecture Specification	Automated Verification Method
CTRL-SEC-01	Algorithmic Cryptographic Transport & Rest Encryption Enforce TLS 1.3 for all in-transit communications and AES-256-GCM for sensitive fields.	Automated security scanning and cryptographic key rotation verification.
CTRL-AUTH-02	Dual-Factor Authentication & Device Fingerprint Binding Mandatory 2FA with device hardware fingerprint binding and session timeout enforcement.	Penetration testing verifying zero session hijack or credential bypass.
CTRL-AUD-03	Statutory Immutable Forensic Audit Logging Maintain append-only WORM audit logs with microsecond NTP timestamps for statutory retention period.	Cryptographic checksum validation & CERT-In audit verification.
CTRL-RES-04	Automated Circuit Breaker & High Availability Failover Deploy Resilience4j circuit breakers and active-standby database failover with RPO < 1m.	Live chaos drill and automated failover switchover test.

5. Traceability Matrix & Recommended SDLC Sprints

Req IDs	Target Sub-Module	SDLC Phase / Gate	Verification Deliverable
REQ-FE-01 - REQ-FE-05	Algorithmic Trading Execution Engine — Frontend & Mobile UI/UX	Sprint 1 / Frontend Milestone	Frontend & Mobile UI/UX Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-BE-01 - REQ-BE-05	Algorithmic Trading Execution Engine — Backend Architecture & Trading Engine	Sprint 2 / Backend Milestone	Backend Architecture & Trading Engine Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-AUTH-01 - REQ-AUTH-05	Algorithmic Trading Execution Engine — Authentication, Identity & Access Governance	Sprint 3 / Authentication, Milestone	Authentication, Identity & Access Governance Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-SEC-01 - REQ-SEC-07	Algorithmic Trading Execution Engine — Information Security & Statutory Compliance	Sprint 4 / Information Milestone	Information Security & Statutory Compliance Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-DB-01 - REQ-DB-05	Algorithmic Trading Execution Engine — Database Architecture & Data Residency	Sprint 5 / Database Milestone	Database Architecture & Data Residency Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-RES-01 - REQ-RES-07	Algorithmic Trading Execution Engine — Resilience, Scalability & Disaster Recovery	Sprint 6 / Resilience, Milestone	Resilience, Scalability & Disaster Recovery Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>

REQ-EXT-35	Algorithmic Trading Execution Engine — Core Features	Sprint 7 / Core Milestone	Core Features Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-36	Algorithmic Trading Execution Engine — Integration	Sprint 8 / Integration Milestone	Integration Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-37	Algorithmic Trading Execution Engine — Testing	Sprint 9 / Testing Milestone	Testing Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-38	Algorithmic Trading Execution Engine — Monitoring	Sprint 10 / Monitoring Milestone	Monitoring Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>